Problem 1: Z-Test (One sample)

Given:

$n = 36$ ✓ $\bar{x} = 495\text{mL}$ $\mu = 500\text{mL}$ $\sigma = 12\text{mL}$ $\alpha = 0.05$ ✓ 5pts

Hypotheses:

H_0: $\mu = 500$

H_a: $\mu \neq 500$ (two-tailed test) ✓ 1pt

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Formula:

$$z = \frac{(\bar{x} - \mu)}{\sigma/\sqrt{n}}$$

Computation/Solution:

$$z = \frac{(495 - 500)}{12/\sqrt{36}}$$

$$z = \frac{-5}{(12/\sqrt{36})}$$ ✓ 1pt

$$z = \frac{-5}{2}$$

$$\boxed{z = -2.50}$$ ✓ 1pt

Decision Rule: 1pt

z critical value $= \pm 1.96$ ✓

$|z| = 2.50$

$2.50 > 1.96 \rightarrow$ Reject H_0 ✓ 1pt

Problem 2: T-test One Sample

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Given:

$n = 25$ ✓

$\bar{x} = 4.2$ hrs ✓

$\mu = 5$ hrs ✓

$s = 1.8$ hrs ✓

$df = N - 1 = 25 - 1 = 24$ ✓ 4pts

$\alpha = 0.05$ ✓

Hypotheses

$H_0 : \mu = 5$

$H_a : \mu \neq 5$ ✓ 1pt

Formula:

$$t = \frac{(\bar{x} - \mu)}{(s/\sqrt{n})}$$

Decision Rule:

t-critical $= \pm 2.064$ ✓ 1pt

$|t| = 2.22$

$2.22 > 2.064 \rightarrow$ Reject H_0 ✓ 1pt

Solution:

$$t = \frac{(4.2 - 5)}{(1.8/\sqrt{25})}$$

$$t = \frac{-0.8}{(1.8/5)}$$ ✓ 1pt

$$t = \frac{-0.8}{0.36}$$

$$t = -2.22$$ ✓ 1pt

Problem 3: One-Way ANOVA

Data	Method A	B	C	
	78	85	90	T = total
	80	88	92	
	82	86	91	
	$T_1 = 240$	$T_2 = 259$	$T_3 = 273$	$= 772$
	$\bar{x} = 80$	$\bar{x} = 86.33$	$\bar{x} = 91$	

Squared Values

A^2	B^2	C^2
6084	7,225	8,100
6,400	7,744	8,464
6,724	7,396	8,281
19,208	22,365	24,845

Totals:

Grand Total = 772

$n = 9$

$\Sigma x^2 = 66,418$

Hypotheses:

$H_0: \mu_1 = \mu_2 = \mu_3$

H_a: Atleast one mean differs

Computation:

Step 1: Compute SST (Total Sum of Squares)

$$SS_{Total} = \Sigma X^2 - \frac{G^2}{N}$$

$$= 66,418 - \frac{772^2}{9}$$

$$= 66,418 - \frac{596,284}{9}$$

$$= 66,418 - 66,220.44$$

$$\boxed{SST = 197.56}$$

Step 2: Compute SS_B (Between-Groups Sum of Squares)

$$SS_B = \frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \frac{T_3^2}{n_3} - \frac{G^2}{N}$$

$$SS_B = \frac{240^2}{3} + \frac{259^2}{3} + \frac{273^2}{3} - \frac{772^2}{9}$$

$$SS_B = \frac{57600}{3} + \frac{67,081}{3} + \frac{74529}{3} - \frac{596,284}{9}$$

$$SS_B = 19200 + 22860.33 + 24.843 - 66220.44$$

$$SS_B = 66403.33 - 66220.44$$

$$\boxed{SS_B = 182.89}$$

Computation:

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$$\boxed{SS_B = 182.89}$$

Step 3: Compute SSW (within)

$SSW = SST - SSB$

$= 197.56 - 182.89$

$\boxed{SSW = 14.67}$

Step 4: Degrees of freedom

$df_B = K-1 \quad = 3-1 = 2$

$df_W = N-K \quad = 9-3 = 6$

$df_t = N-1 \quad = 9-1 = 8$

Step 5: Compute MS (Mean Squares)

$MSB = \frac{SSB}{df_B} = \frac{182.89}{2}$

$\boxed{= 91.45}$

$MSW = \frac{SSW}{df_W} = \frac{14.67}{6}$

$\boxed{MSW = 2.44}$

Step 6: Compute F-Ratio:

$F = \frac{MSB}{MSW} = \frac{91.45}{2.44}$

$\boxed{F = 37.48}$

ANOVA Table (Summary)

Source	SS	df	MS	F
Between	182.89	2	91.45	37.48
Within	14.67	6	2.44	—
Total	197.56	8	—	—

Decision Rule:

$F_{critical}(2,6) = 5.14$

$37.40 > 5.14 \rightarrow$ Reject H_0 ✓

Problem 4: Pearson Correlation

table

X	Y	X^2	Y^2	XY
2	45	4	4,225	130
3	70	9	4,900	210
4	75	16	5,625	300
5	80	25	6,400	400
6	85	36	7,225	510
$\Sigma = 20$	$\Sigma = 375$	$\Sigma = 90$	$\Sigma = 28,375$	$\Sigma = 1,550$

Hypotheses:

$H_0: r = 0$

$H_a: r \neq 0$

Formula:

$$r = \frac{n\Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{[n\Sigma X^2 - (\Sigma X)^2][n\Sigma Y^2 - (\Sigma Y)^2]}}$$

Computation:

$$r = \frac{5(1,550) - (20)(375)}{\sqrt{[5(90) - 20^2][5(28,375) - 375^2]}}$$

$450 - 400 \qquad 141,875 \quad 140,625$

$$r = \frac{250}{250}$$

$$\boxed{r = 1.00}$$ Perfect Correlation

Problem 5

X	X	X^2	X^2	XX
95	80	9025	6,400	7600
100	83	10,000	6889	8300
105	85	11,025	7225	8925
110	89	12,100	7921	9790
115	92	13,225	8464	10580
Σ=525	Σ=429	Σ=55,375	Σ=36,899	Σ=45195

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Hypotheses:

H_0 : $r = 0$

H_a : $r \neq 0$

formula:

$$r = \frac{n\Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{[n\Sigma X^2 - (\Sigma X)^2][n\Sigma Y^2 - (\Sigma Y)^2]}}$$

Computation:

$$r = \frac{5(45,195) - (525)(429)}{\sqrt{[5(55,375) - 525^2][5(36,899) - 429^2]}}$$

$$r = \frac{750}{753.33} = 0.996$$

$r = 0.996$ Very high positive correlation

problem 6:

x	y	x^2	y^2
10	100	100	1,000
15	130	225	1,950
20	150	400	3,000
20	180	625	4,500
25	210	900	6,300
$\Sigma = 100$	$\Sigma = 770$	$\Sigma = 2,250$	$\Sigma = 16,750$

6 pt

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Slope (b):

formula:

$$b = \frac{n\Sigma XY - (\Sigma X)(\Sigma Y)}{n\Sigma X^2 - (\Sigma X)^2}$$

$$b = \frac{5(16750) - (100)(770)}{5(2250) - (100)^2}$$

$$= \frac{83,750 - 77000}{11,250 - 10,000}$$

$$b = \frac{6750}{1250}$$

$$b = 5.4$$

Intercept (a)

$$a = \frac{(\Sigma Y)(\Sigma X^2) - (\Sigma X)(\Sigma XY)}{n\Sigma X^2 - (\Sigma X)^2}$$

$$a = \frac{(770)(2250) - (100)(16750)}{1250}$$

$$a = \frac{1,732,500 - 1675000}{1250}$$

$$a = \frac{57500}{1250}$$

$$a = 46$$

Regression equation = $Y = 46 + 5.4X$

Prediction when $X = 50$

$$Y = 46 + 5.4(50)$$

$$= 46 + 270$$

$$Y = 316$$